卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-CET-DD-EC-0213 49-B60011S-D0213

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气一次 专业 第 02 卷 第 13 册
Electric SPECIALITY VOLUME BOOK

卷册名称 132kV交流配电装置接线及平、断面图
Volume name Single Line Diagram and Arrangement of AC 132kV Switchgear

批 准 APPROVED BY	邓伟
审 核 REVIEWED BY	田俊强
设计/勘测 DESIGNED BY/ SURVEYED BY	葛斌
图 纸 5 张 DRAWING NOS.	打印件 / 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
01	49-B60011S-D0201-001	卷册说明 Description of Volume	A	1		
02	49-B60011S-D0201-002	132kV交流配电装置接线及平面布置图 132kV Switchyard Single Line and Layout Plan	A	1		
03	49-B60011S-D0201-003	132kV交流配电装置断面图(一) AC 132kV Switchgear Section (1)	A	1		
04	49-B60011S-D0201-004	132kV交流配电装置断面图(二) AC 132kV Switchgear Section (2)	A	1		
05	49-B60011S-D0201-005	主要设备材料汇总表 Summary of Major Equipment and Materials	A	1		

注:

Description of Volume

1. This volume include single line diagram and arrangement of AC 132kV switchgear.
   The boundary between this volume and Outline is the strain insulator strings at 132kV switchyard side. The clamp and connection conductors are listed in this volume. The strain insulator string and outline conductors are not in this volume.
2. Suspended ø170/154 Pipe buses made of 6063G Aluminum-Magnesium are applied as main bus of AC 132 kV AC switchgear; The conductor is Stranded Aluminum Wire JL-800-61.
3. When the foundations are under construction, installing workers should cooperate with the civil workers and check the equipment that arrives at the substation to make sure it is in line with design diagram.
4. Installing workers must check the foundations and the supports are in line with the design diagram.
5. Sags of connection lines between equipment are set as 0.3m~0.7m. When installing the connection lines, pay attentions to ensure tension force should not exceed the maximum load of the equipment's terminal.
6. According to IEC standard, the minimum air clearance in this project are list as the following table:

Code \ Voltage Level	132kV
minimum clearance of phase to earth	1300mm (N)
minimum clearance of different phase	1300mm (N)
minimum clearance between live part with transport	1400mm (N+100)
minimum clearance between live part with meshes	1380mm (N+80)
minimum clearance between live conductor with earth	3550mm (N+2250)
minimum clearance(horizontal) between outdoor live conductor with unscreened house	3300mm (N+2000)

7. Grounding Conductor and fittings are shown in "49-B60011S-D0403".
8. The clamps in this volume is listed by Pinggao Group Co., Ltd. Clamp terminal's size are designed as length x width.

卷册说明

1. 本分册范围包括132kV交流配电装置的接线和平、断面布置。
   本卷册与线路出线的分界点在出线耐张串。引下线夹、跳线及引下线由本册开列，耐张串及线路导线不在本卷册设计范围。
2. 132kV交流配电装置主母线采用6063G-ø170/154管母；导线采用JL-800-61纯铝绞线。
3. 土建施工设备基础时，电气安装人员应密切配合，并应核对现场实际到货设备是否与设计图纸相符。
4. 电气施工前，电气安装人员应核对设备基础或支架高度及定位是否与设计图纸相符。
5. 设备间连线弧垂为0.3~0.7m。安装设备间连接导线时，导线拉力不应超过设备接线端子所允许的受力要求。
6. 本工程空气净距根据IEC标准选取，户外配电装置的最小空气净距列于下表中：

类别 \ 电压等级	132kV
相对地最小距离	1300mm (N)
相间最小距离	1300mm (N)
对交通工具的最小距离	1400mm (N+100)
对网格围栏的最小距离	1380mm (N+80)
裸导体距地面最小距离	3550mm (N+2250)
户外导线对有窗户的建筑物的最小距离	3300mm (N+2000)

7. 避雷线及相关金具开列在"40-BA08491S- D0403"中。
8. 金具型号由平高集团股份有限公司开列。金具接线端子板的尺寸按长x宽开列。

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

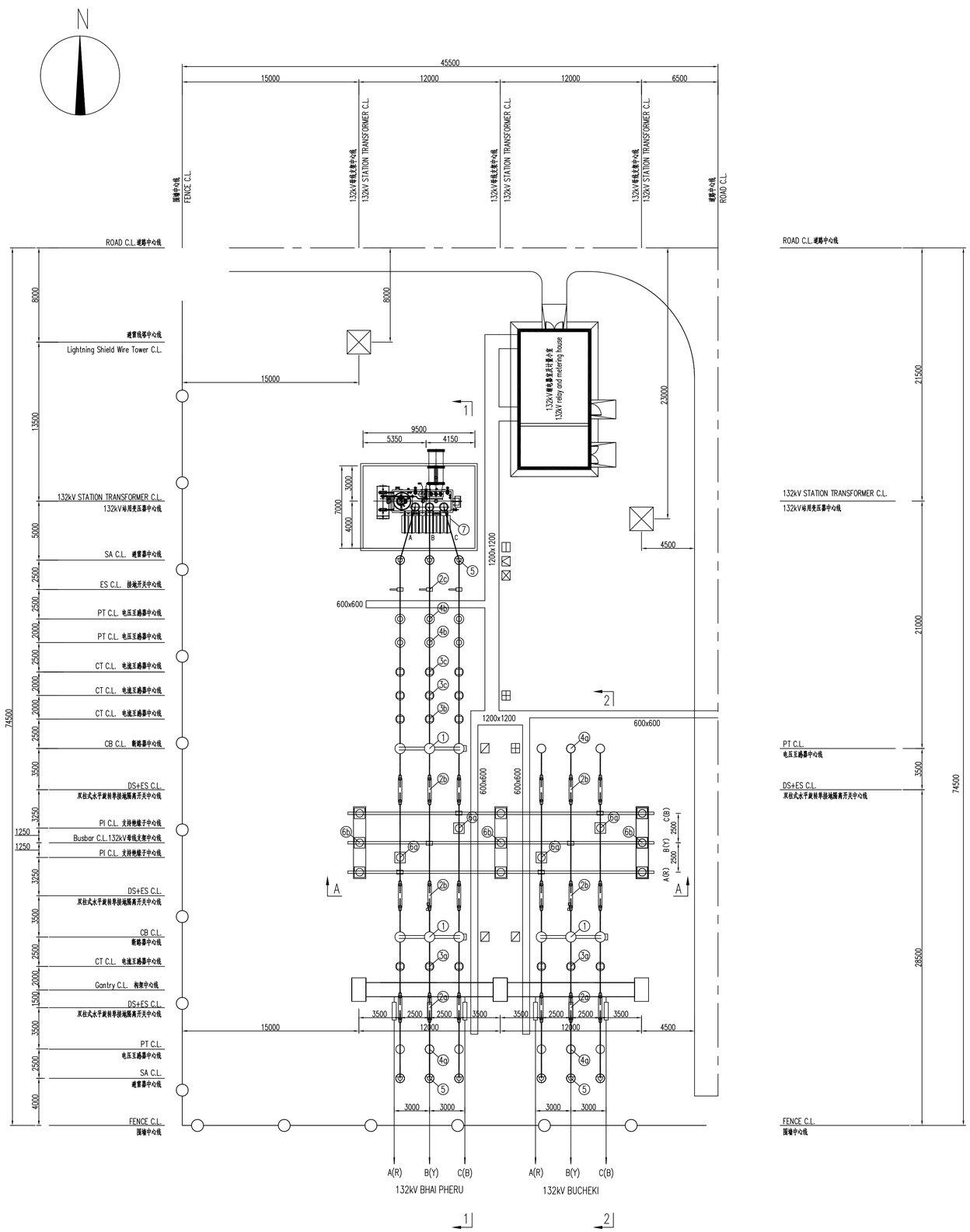
M L H V D C - L C S - C E T - D D - E C - 0 2 1 3 - 0 0 1 A

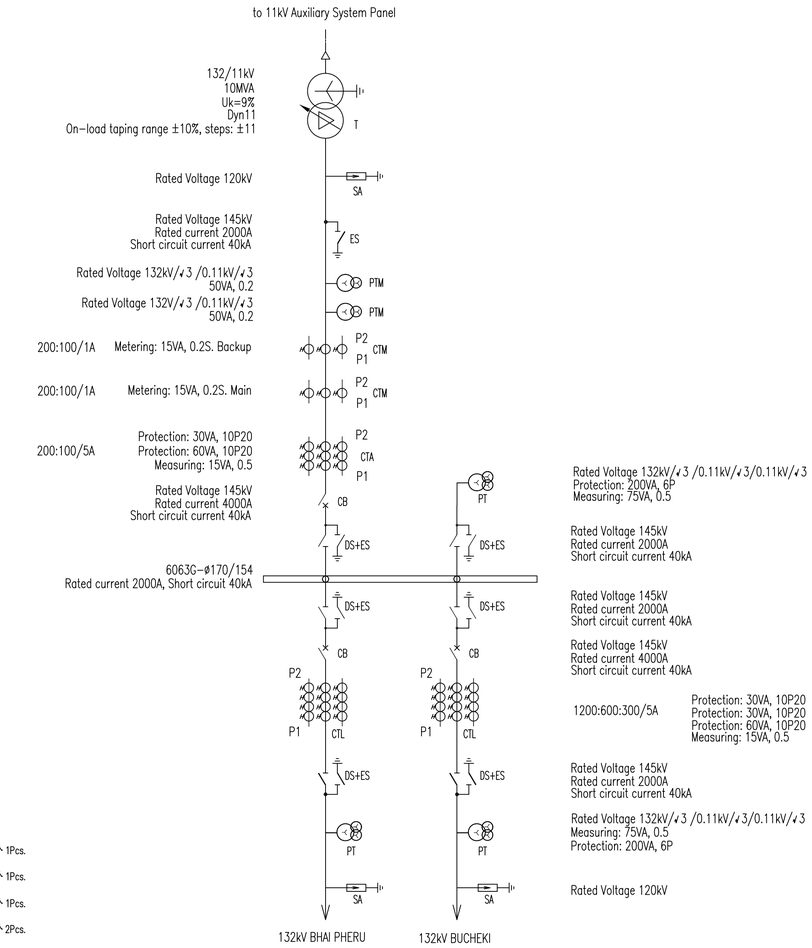
	姓名 Name	日期 Date
设计 Designed	葛斌	2019.09.20
校核 Checked	李远	2019.09.20
审核 Reviewed	田俊强	2019.09.20
批准 Approved	刘伟	2019.09.20

卷册说明
Description of Volume

比例 Scale: 1:1
图号 DWG NO.: 49-B60011S-D0213-001
版本 REV.: A

297X420





图例 Legend:

检修箱 Maintenance box	⊟	1个 1Pcs.
交流电源配电箱 AC power switch box	⊠	1个 1Pcs.
直流电源配电箱 DC power switch box	☑	1个 1Pcs.
CVT端子箱 CVT terminal box	⊞	2个 2Pcs.
断路器端子箱/间隔工控箱 CB terminal box	◪	3个 3Pcs.

132kV区域设备表

Equipment table of 132kV Switchyard

编号 No.		设备名称 Equipment	设备参数 Parameters of the Equipment	单位 Unit	数量 QYT.	厂家 Manufacturer
5	SA	避雷器 Surge Arrester	Rated Voltage 120kV	台 Set	9	扶申电瓷制造有限公司 Fushun electric porcelain manufacturing co., ltd.
4a	PT	电压互感器 Voltage transformer	132kV/√3 /0.11kV/√3/0.11kV/√3 Measuring: 75VA, 0.5 Protection: 200VA, 6P	台 Set	9	阿特克大一互 ARTECHE DYH ELECTRIC CO., LTD.
4b	PTM	计量电压互感器 Metering Voltage transformer	132V/√3 /0.11kV/√3 50VA, 0.2	台 Set	6	阿特克大一互 ARTECHE DYH ELECTRIC CO., LTD.
3a	CTL	出线电流互感器 Line current transformer	1200:600:300/5A, Protection: 30VA, 10P20 Protection: 30VA, 10P20 Protection: 60VA, 10P20 Measuring: 15VA, 0.5	台 Set	6	阿特克大一互 ARTECHE DYH ELECTRIC CO., LTD.
3b	CTA	站用间隔电流互感器 Current transformer for auxiliary transformer	200: 100/5A, Measuring: 15VA, 0.5 Protection: 60VA, 10P20 Protection: 30VA, 10P20	台 Set	3	阿特克大一互 ARTECHE DYH ELECTRIC CO., LTD.
3c	CTM	站用间隔电流互感器 Current transformer for auxiliary transformer	200: 100/1A, Metering: 15VA, 0.2S. Main Metering: 15VA, 0.2S. Backup	台 Set	6	阿特克大一互 ARTECHE DYH ELECTRIC CO., LTD.
2a、2b	DS	隔离开关 Disconnect Switch	Rated Voltage 145kV, rated current 2000A Short circuit current 40kA	组 Set	6	江苏如高高压电器有限公司 Jiangsu Ruigao High Voltage Electric Apparatus Co., Ltd.
2c	ES	接地开关 Earthing Switch	Rated Voltage 145kV, Short circuit current 40kA	组 Set	1	江苏如高高压电器有限公司 Jiangsu Ruigao High Voltage Electric Apparatus Co., Ltd.
1	CB	断路器 Breaker	Rated Voltage 145kV, rated current 4000A Short circuit current 40kA	台 Set	3	江苏如高高压电器有限公司 Jiangsu Ruigao High Voltage Electric Apparatus Co., Ltd.
6a、6b	PI	支柱绝缘子 Post Insulator	Rated phase-to-ground Voltage 132kV	支 Set	13	扶申电瓷制造有限公司 Fushun electric porcelain manufacturing co., ltd.
7	T	132kV站用变压器 132kV Auxiliary transformer	132/11kV, 10MVA, Uk=9%, Dyn11 On-load taping range ±10%, steps: ±11	台 Set	1	山东电力设备有限公司

说明:

1.图中尺寸单位为mm。

Note:

1.All units of dimension in this drawing are mm.

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中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT & ENGINEERING CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 1 3 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	高士成	2019.10.31
校核 Checked	[illegible]	2019.10.31
审核 Reviewed	田俊辉	2019.10.31
批准 Approved	[illegible]	2019.10.31

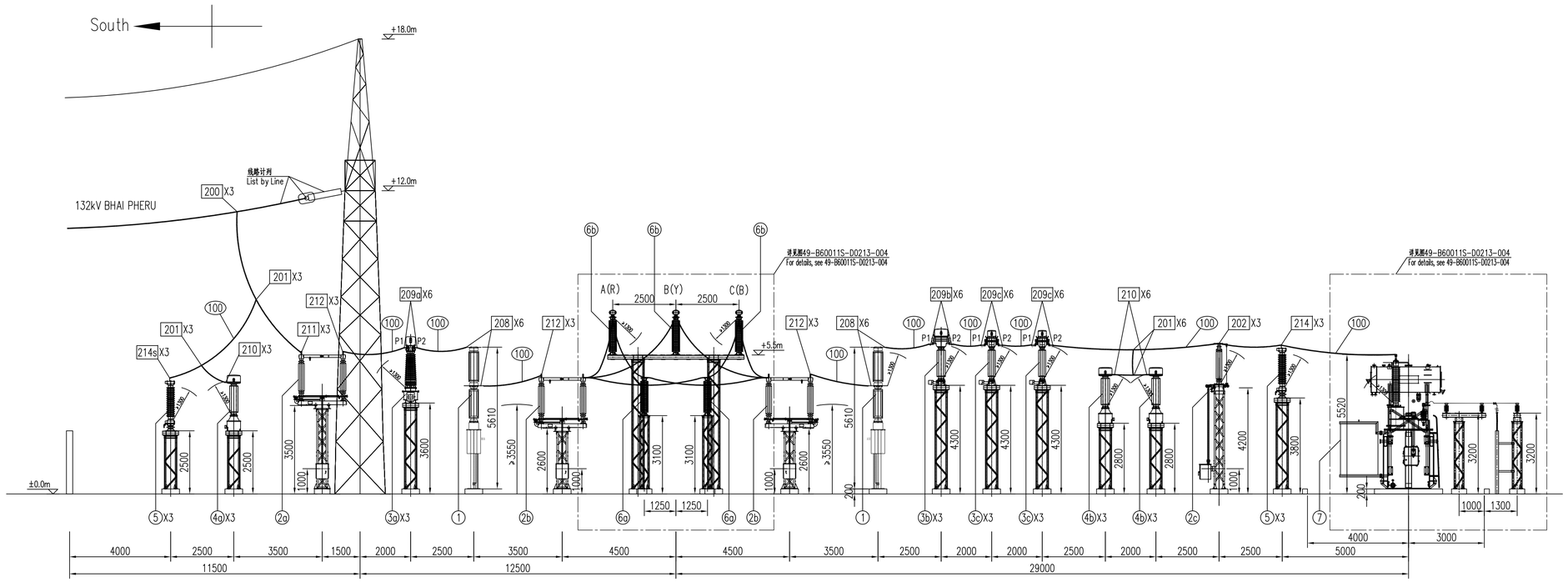
结构	[illegible]	2019.10
专业 Speciality	会签 Sign	日期 Date

132kV配电装置接线及平面布置图

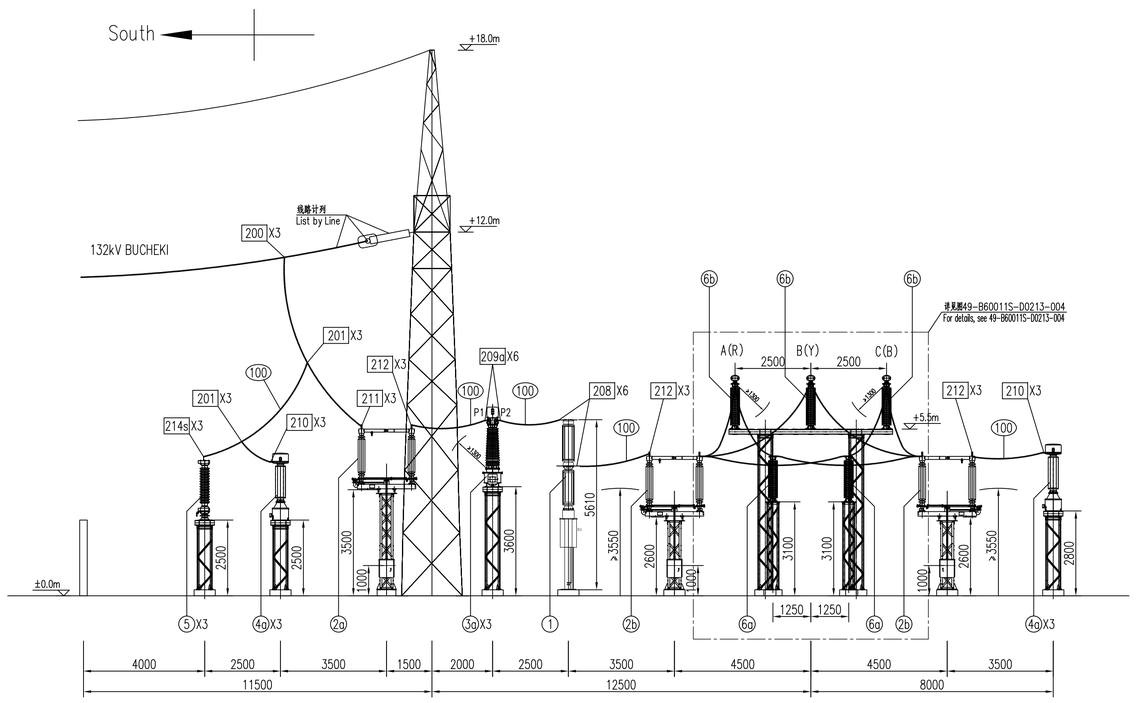
132kV Switchyard Single Line and Layout Plan

比例 Scale: 1:200 图号 DWG NO.: 49-B60011S-D0213-002 版本 REV.: A

594X841



1-1



2-2

214s	45°单导线铝设备线夹	SY-800B-SA(120X110)	套	6	甲供
214	T型线夹	TLL-800B-SA(120X110)	套	3	甲供
212	90°单导线铝设备线夹	SYB-800C-DS(130X⌀40)	套	18	甲供
211	45°单导线铝设备线夹	SYB-800B-DS(130X⌀40)	套	6	甲供
210	0° 单导线铝设备线夹	SYB-800A-PT(80X⌀30)，镀银	套	15	甲供
209c	0° 单导线铝设备线夹	SYB-800A-CT(125X⌀60)，镀银	套	12	甲供，用于站用变计量CT
209b	0° 单导线铝设备线夹	SYB-800A-CT(125X⌀30)，镀银	套	6	甲供，用于站用变回路CT
209a	0° 单导线铝设备线夹	SYB-800A-CT(125X⌀40)，镀银	套	12	甲供，用于出线回路CT
208	0° 单导线铝设备线夹	SY-800A-CB(110x150)	套	18	甲供
202	软导线固定金具	MDG-800-ES(⌀140-4X⌀14)	套	3	甲供
201	T型线夹	TLY-800/800	套	18	甲供
200	T型线夹	TLY-Rail/800	套	6	甲供
100	纯铝绞线	JL-800-61	米	300	甲供
编号	名　称	型号及规范	单位	数量	备　注

设备材料表

214s	Compression Terminds for Single Conductor (Type SYB,Group B,45°)	SY-800B-SA(120X110)	Set	6	Supplied by Owner
214	T-Connector(Type TLL)	TLL-800B-SA(120X110)	Set	3	Supplied by Owner
212	Compression Terminds for Single Conductor (Type SYB,Group C,90°)	SYB-800C-DS(130X⌀40)	Set	18	Supplied by Owner
211	Compression Terminds for Single Conductor (Type SYB,Group B,45°)	SYB-800B-DS(130X⌀40)	Set	6	Supplied by Owner
210	Compression Terminds for Single Conductor (Type SYB,Group A,0°)	SYB-800A-PT(80X⌀30),Silvering	Set	15	Supplied by Owner
209c	Compression Terminds for Single Conductor (Type SYB,Group A,0°)	SYB-800A-CT(125X⌀60),Silvering	Set	12	Supplied by Owner
209b	Compression Terminds for Single Conductor (Type SYB,Group A,0°)	SYB-800A-CT(125X⌀30),Silvering	Set	6	Supplied by Owner
209a	Compression Terminds for Single Conductor (Type SYB,Group A,0°)	SYB-800A-CT(125X⌀40),Silvering	Set	12	Supplied by Owner
208	Compression Terminds for Single Conductor (Type SY,Group A,0°)	SY-800A-CB(110x150)	Set	18	Supplied by Owner
202	Single Bus Support(Type MDG)	MDG-800-ES(⌀140-4X⌀14)	Set	3	Supplied by Owner
201	T-Connector(Type TLY)	TLY-800/800	Set	18	Supplied by Owner
200	T-Connector(Type TLY)	TLY-Rail/800	Set	6	Supplied by Owner
100	Stranded Aluminum Wire	JL-800-61	m	300	Supplied by Owner
No.	Name	Type and Details	Unit	Qty.	Remark

Table of Equipment and Materials

说明：
1.图中尺寸单位为mm。

Note:
1.All units of dimension in this drawing are mm.

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CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 1 3 - 0 0 3 A

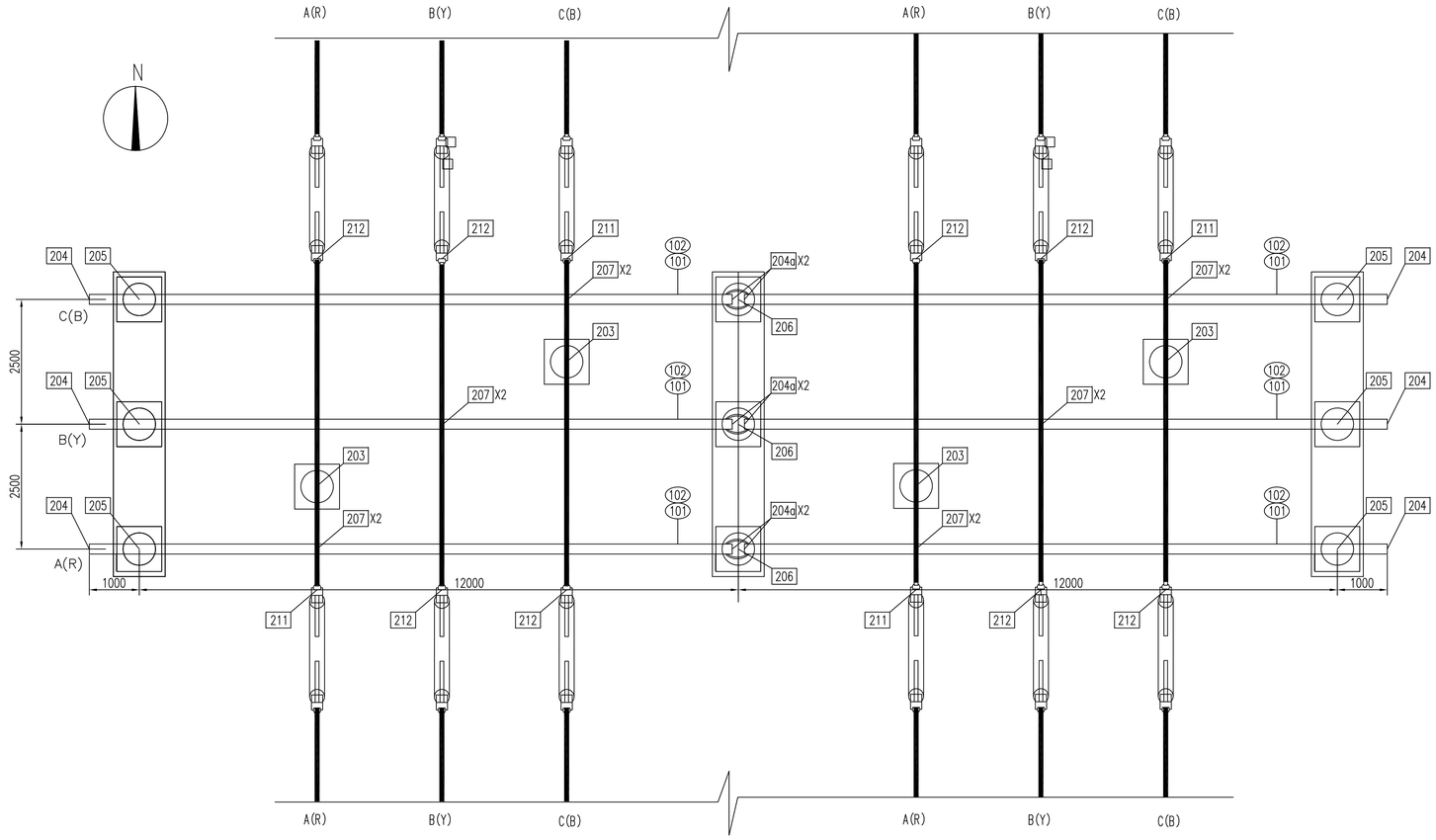
	姓名 Name	日期 Date
设计 Designed	苟文斌	2019.12.20
校核 Checked	[illegible]	2019.12.20
审核 Reviewed	田俊强	2019.12.20
批准 Approved	[illegible]	2019.12.20
比例 Scale	1:100	图号 DWG NO. 49-B60011S-D0213-003

132kV交流配电装置断面图（一）
AC 132kV Switchgear Section (1)

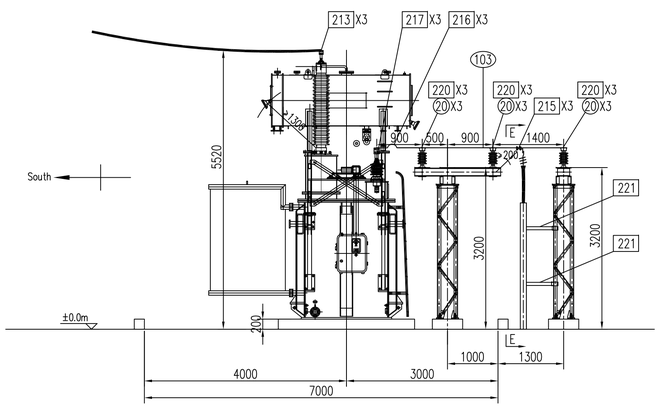
版本 REV. A

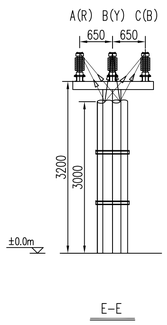
结构	[illegible]	2019.12
专业 Speciality	会签 Sign	日期 Date

594X841



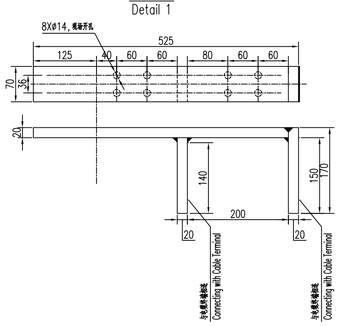
132kV母线区域俯视图
The layout plan of the 132kV bus switchyard



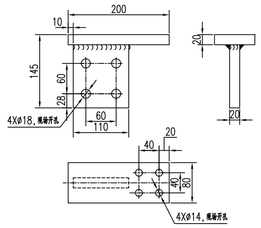


E-E

详图1
Detail 1



详图2
Detail 2



说明:
1.图中尺寸单位为mm。

Note:
1.All units of dimension in this drawing are mm.

No.	Name	Type and Details	Unit	Qty.	Remark
221	Hoop	On-site Processing	Set	2	Supplied by Constructor
220	Bus Support(Type MWP)	MWP-102	Set	9	Supplied by Owner
20	Post Insulator	ZS-20/30	Pcs.	9	Supplied by Constructor
217	110-80 90° Adaptor Plate	200X80 (Special Clamp)	Set	3	Supplied by Owner See Detail 2
216	Flexible Laminated Copper Connector (Type MST)	SYB-800C-T(80X80)	Set	3	Supplied by Owner
215	Copper Transition Plate	525X70 (Special Clamp)	Set	3	Supplied by Owner See Detail 1
213	Cu-Al Compression Terminals for Single Conductor (Type SYG,Group A,0°)	SYG-800A-T(80X80)	Set	3	Supplied by Owner
212	Compression Terminals for Single Conductor (Type SYB,Group C,90°)	SYB-800C-DS(130X∅40)	Set	8	Supplied by Owner
211	Compression Terminals for Single Conductor (Type SYB,Group B,45°)	SYB-800B-DS(130X∅40)	Set	4	Supplied by Owner
207	T-Connector(Type MGTY)	MGTY-∅170-800	Set	12	Supplied by Owner
206	Flexible Bus Support(Type MGSZ)	MGSZ-170(∅225-4X∅18)	Set	3	Supplied by Owner
205	Fixed Bus Support(Type MGG2)	MGG2-170(∅225-4X∅18)	Set	6	Supplied by Owner
204a	End-Caps(Type MGF)	MGF-170	Set	6	Supplied by Owner
204	End-Caps(Type MGF1)	MGF1-170	Set	6	Supplied by Owner
203	Single Bus Support(Type MDG)	MDG-800-PI(∅225-4X∅18)	Set	4	Supplied by Owner
103	Flat Copper	80X10,L=5m	Pcs	3	Supplied by Owner
102	Aluminum Alloy Tube	6063G-∅170/154,L=13m	Pcs	6	Supplied by Owner
101	ASCR	JL/G1A-400/35-48/7	m	85	Supplied by Owner

Table of Equipment and Materials

编号	名 称	型号及规范	单位	数量	备 注
221	抱箍	现场加工	套	2	乙供
220	母线固定金具	MWP-102	套	9	甲供
20	支柱绝缘子	ZS-20/30,顶部开孔匹配MWP-102	个	9	乙供
217	110-80 90°转接板	200X80 (特制金具)	套	3	甲供,见详图2
216	全铜母线伸缩节	MST-80X8	套	3	甲供
215	90°全铜过渡板	525X70 (特制金具)	套	3	甲供,见详图1
213	0°单导线铜铝过渡设备线夹	SYG-800A-T(80X80)	套	3	甲供
212	90°单导线铝设备线夹	SYB-800C-DS(130X∅40)	套	8	甲供
211	45°单导线铝设备线夹	SYB-800B-DS(130X∅40)	套	4	甲供
207	管母单引流线夹	MGTY-∅170-800	套	12	甲供
206	管母伸缩线夹	MGSZ-170(∅225-4X∅18)	套	3	甲供
205	管母固定线夹	MGG2-170(∅225-4X∅18)	套	6	甲供
204a	管母封端盖	MGF-170	套	6	甲供
204	管母阻尼线封端盖	MGF1-170	套	6	甲供
203	软导线固定金具	MDG-800-PI(∅225-4X∅18)	套	4	甲供
103	镀锡铜排	80X10,L=5m	根	3	甲供
102	铝镁硅合金管母	6063G-∅170/154,L=13m	根	6	甲供
101	钢芯铝绞线	JL/G1A-400/35-48/7	米	85	甲供,用于阻尼线

设备材料表

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR
国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 1 3 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	高旭	2019.12.20
校核 Checked		2019.12.20
审核 Reviewed	田俊强	2019.12.20
批准 Approved		2019.12.20
比例 Scale	1:60	图号 DWG NO. 49-B60011S-D0213-004

132kV交流配电装置断面图(二)
AC 132kV Switchgear Section (2)

版本 REV. A

专业 Speciality	会签 Sign	日期 Date
结构		2019.12

594X841

编号	名　称	型号及规范	单位	数量	备　注
221	抱箍	现场加工	套	2	乙供
220	母线固定金具	MWP-102	套	9	甲供
20	支柱绝缘子	ZS-20/30，顶部开孔匹配MWP-102	个	9	乙供
217	110-80 90°转接板	200X80（特制金具）	套	3	甲供
216	全铜母线伸缩节	MST-80X8	套	3	甲供
215	90°全铜过渡板	525X70（特制金具）	套	3	甲供
214s	45°单导线铝设备线夹	SY-800B-SA(120X110)	套	6	甲供
214	T型线夹	TLL-800B-SA(120X110)	套	3	甲供
213	0° 单导线铜铝过渡设备线夹	SYG-800A-T(80X80)	套	3	甲供
212	90°单导线铝设备线夹	SYB-800C-DS(130X∅40)	套	26	甲供
211	45°单导线铝设备线夹	SYB-800B-DS(130X∅40)	套	10	甲供
210	0° 单导线铝设备线夹	SYB-800A-PT(80X∅30)，镀银	套	15	甲供
209c	0° 单导线铝设备线夹	SYB-800A-CT(125X∅60)，镀银	套	12	甲供，用于站用变计量CT
209b	0° 单导线铝设备线夹	SYB-800A-CT(125X∅30)，镀银	套	6	甲供，用于站用变回路CT
209a	0° 单导线铝设备线夹	SYB-800A-CT(125X∅40)，镀银	套	12	甲供，用于出线回路CT
208	0° 单导线铝设备线夹	SY-800A-CB(110x150)	套	18	甲供
207	管母单引流线夹	MGTY-∅170-800	套	12	甲供
206	管母伸缩线夹	MGSZ-170(∅225-4X∅18)	套	3	甲供
205	管母固定线夹	MGG2-170(∅225-4X∅18)	套	6	甲供
204a	管母封端盖	MGF-170	套	6	甲供
204	管母阻尼线封端盖	MGF1-170	套	6	甲供
203	软导线固定金具	MDG-800-PI(∅225-4X∅18)	套	4	甲供
202	软导线固定金具	MDG-800-ES(∅140-4X∅14)	套	3	甲供
201	T型线夹	TLY-800/800	套	18	甲供
200	T型线夹	TLY-Rail/800	套	6	甲供
103	镀锡铜排	80X10，L=5m	根	3	甲供
102	铝镁硅合金管母	6063G-∅170/154，L=13m	根	6	甲供
101	钢芯铝绞线	JL/G1A-400/35-48/7	米	85	甲供，用于阻尼线
100	纯铝绞线	JL-800-61	米	300	甲供

设备材料表

No.	Name	Type and Details	Unit	Qty.	Remark
221	Hoop	On-site Processing	Set	2	Supplied by Constructor
220	Bus Support(Type MWP)	MWP-102	Set	9	Supplied by Owner
20	Post Insulator	ZS-20/30	Pcs.	9	Supplied by Constructor
217	110-80 90° Adaptor Plate	200X80 (Special Clamp)	Set	3	Supplied by Owner
216	Flexible Laminated Copper Connector (Type MST)	SYB-800C-T(80X80)	Set	3	Supplied by Owner
215	Copper Transition Plate	525X70 (Special Clamp)	Set	3	Supplied by Owner
214s	Compression Terminals for Single Conductor (Type SY,Group B,45°)	SY-800B-SA(120X110)	Set	6	Supplied by Owner
214	T-Connector(Type TLL)	TLL-800B-SA(120X110)	Set	3	Supplied by Owner
213	Cu-Al Compression Terminals for Single Conductor (Type SYG,Group A,0°)	SYG-800A-T(80X80)	Set	3	Supplied by Owner
212	Compression Terminals for Single Conductor (Type SYB,Group C,90°)	SYB-800C-DS(130X∅40)	Set	26	Supplied by Owner
211	Compression Terminals for Single Conductor (Type SYB,Group B,45°)	SYB-800B-DS(130X∅40)	Set	10	Supplied by Owner
210	Compression Terminals for Single Conductor (Type SYB,Group A,0°)	SYB-800A-PT(80X∅30),Silvering	Set	15	Supplied by Owner
209c	Compression Terminals for Single Conductor (Type SYB,Group A,0°)	SYB-800A-CT(125X∅60),Silvering	Set	12	Supplied by Owner
209b	Compression Terminals for Single Conductor (Type SYB,Group A,0°)	SYB-800A-CT(125X∅30),Silvering	Set	6	Supplied by Owner
209a	Compression Terminals for Single Conductor (Type SYB,Group A,0°)	SYB-800A-CT(125X∅40),Silvering	Set	12	Supplied by Owner
208	Compression Terminals for Single Conductor (Type SY,Group A,0°)	SY-800A-CB(110x150)	Set	18	Supplied by Owner
207	T-Connector(Type MGTY)	MGTY-∅170-800	Set	12	Supplied by Owner
206	Flexible Bus Support(Type MGSZ)	MGSZ-170(∅225-4X∅18)	Set	3	Supplied by Owner
205	Fixed Bus Support(Type MGG2)	MGG2-170(∅225-4X∅18)	Set	6	Supplied by Owner
204a	End-Caps(Type MGF)	MGF-170	Set	6	Supplied by Owner
204	End-Caps(Type MGF1)	MGF1-170	Set	6	Supplied by Owner
203	Single Bus Support(Type MDG)	MDG-800-PI(∅225-4X∅18)	Set	4	Supplied by Owner
202	Single Bus Support(Type MDG)	MDG-800-ES(∅140-4X∅14)	Set	3	Supplied by Owner
201	T-Connector(Type TLY)	TLY-800/800	Set	18	Supplied by Owner
200	T-Connector(Type TLY)	TLY-Rail/800	Set	6	Supplied by Owner
103	Flat Copper	80X10，L=5m	Pcs	3	Supplied by Owner
102	Aluminum Alloy Tube	6063G-∅170/154，L=13m	Pcs	6	Supplied by Owner
101	ASCR	JL/G1A-400/35-48/7	m	85	Supplied by Owner
100	Stranded Aluminum Wire	JL-800-61	m	300	Supplied by Owner

Table of Equipment and Materials

132kV区域设备表

Equipment table of 132kV Switchyard

编号 No.		设备名称 Equipment	设备参数 Parameters of the Equipment	单位 Unit	数量 QYT.	厂家 Manufacturer
5	SA	避雷器 Surge Arrester	Rated Voltage 120kV	台 Set	9	抚顺电瓷制造有限公司 Fushun electric porcelain manufacturing co., ltd.
4a	PT	电压互感器 Voltage transformer	132kV/√3 /0.11kV/√3/0.11kV/√3 Measuring: 75VA, 0.5 Protection: 200VA, 6P	台 Set	9	阿蒂莫大一互 ARTECHE DYH ELECTRIC CO., LTD.
4b	PTM	计量电压互感器 Metering Voltage transformer	132V/√3 /0.11kV/√3 50VA, 0.2	台 Set	6	阿蒂莫大一互 ARTECHE DYH ELECTRIC CO., LTD.
3a	CTL	出线电流互感器 Line current transformer	1200:600:300/5A, Protection: 30VA, 10P20 Protection: 30VA, 10P20 Protection: 60VA, 10P20 Measuring: 15VA, 0.5	台 Set	6	阿蒂莫大一互 ARTECHE DYH ELECTRIC CO., LTD.
3b	CTA	站用电回路电流互感器 Current transformer for auxiliary transformer	200: 100/5A, Measuring: 15VA, 0.5 Protection: 60VA, 10P20 Protection: 30VA, 10P20	台 Set	3	阿蒂莫大一互 ARTECHE DYH ELECTRIC CO., LTD.
3c	CTM	站用电回路电流互感器 Current transformer for auxiliary transformer	200: 100/1A, Metering: 15VA, 0.2S. Main Metering: 15VA, 0.2S. Backup	台 Set	6	阿蒂莫大一互 ARTECHE DYH ELECTRIC CO., LTD.
2a、2b	DS	隔离开关 Disconnect Switch	Rated Voltage 145kV, rated current 2000A Short circuit current 40kA	组 Set	6	江苏如高高压电器有限公司 Jiangsu Rugao High Voltage Electric Apparatus Co., Ltd.
2c	ES	接地开关 Earthing Switch	Rated Voltage 145kV, Short circuit current 40kA	组 Set	1	江苏如高高压电器有限公司 Jiangsu Rugao High Voltage Electric Apparatus Co., Ltd.
1	CB	断路器 Breaker	Rated Voltage 145kV, rated current 4000A Short circuit current 40kA	台 Set	3	江苏如高高压电器有限公司 Jiangsu Rugao High Voltage Electric Apparatus Co., Ltd.
6a、6b	PI	支柱绝缘子 Post Insulator	Rated phase-to-ground Voltage 132kV	支 Set	13	抚顺电瓷制造有限公司 Fushun electric porcelain manufacturing co., ltd.
7	T	132kV站用变压器 132kV Auxiliary transformer	132/11kV, 10MVA, Uk=9%, Dyn11 On-load taping range ±10%, steps: ±11	台 Set	1	山东电力设备有限公司

132kV区域材料表

Materials table of 132kV Switchyard

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

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设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 1 3 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	葛斌	2019.12.20
校核 Checked	秦迎	2019.12.20
审核 Reviewed	田俊强	2019.12.20
批准 Approved	张伟	2019.12.20

主要设备材料汇总表

Summary of Major Equipment and Materials

比例 Scale: 1:100　图号 DWG NO.: 49-B60011S-D0213-005　版本 REV.: A

420X594